BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Network Topologies

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What does DNS do?

2. Why is a thermostat controller an embedded system?
   A) It has a specific task
   B) It runs any desktop app
   C) It always needs Wi-Fi
   D) It always has a screen

3. What makes a system embedded?

4. A circuit uses two voltage ranges for two states. "Computers use binary because people cannot write numbers above 1." Correct this. Why are two states useful?

Today's objectives

Recognise star and mesh network topologies. Explain advantages and limitations of each.

Key vocabulary - Read, cover, recall

Topology
Arrangement of network devices and links.

Star topology
Endpoints connect to a central switch or hub.

Mesh topology
Devices connect via multiple paths between nodes.

Fault tolerance
Ability to continue working after some faults.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Topology describes connections

A network topology shows how computers and connection devices link together.

Star topology

Each endpoint connects to a central device such as a switch. Easier to add/manage devices, but the central device is a point of failure.

Mesh topology

Devices can connect to multiple others. Alternative paths make the system resilient if one link fails, but more connections increase cost and complexity.

Compare in context

For a small school LAN, a star can be practical. A high-availability network may use mesh routes; explain what happens when a link or central device fails.

I Do - Worked model

A school LAN uses a star topology. Explain one advantage and one disadvantage if its central switch fails. [4 marks]

- All classroom devices connect to one central switch.
- New devices can be connected easily [1], making the network straightforward to manage [1].
- The switch is a single point of failure [1], so its failure can interrupt communication for connected devices [1].
- Give advantage plus linked benefit and disadvantage plus linked effect.

We Do - Guided practice

1. A business needs reliable connections between several sites. Explain one advantage and one disadvantage of a mesh topology. [4 marks]

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2. A small primary school chooses a star topology. Explain two advantages of this decision. [4 marks]

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3. Describe how a mesh can continue operating after one connection fails. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the count of links for a small full mesh.

```
devices = 4
links = devices * (devices - 1) DIV 2
print(links)
```

You Do - Independent exam questions

Q2. State one advantage of a mesh topology. [1 mark]

Q3. A cinema uses a star network to connect ticket kiosks and staff computers. The central switch stops working. Explain the effect and suggest one precaution. [4 marks]

Q4. Compare star and mesh topologies for a large organisation. [6 marks]

Exit Ticket - True or False?

Mesh topology generally provides alternative links that can improve fault tolerance.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. Distributed DNS servers help look up the IP address associated with a domain name.

2. A - It has a specific task

3. It is built into a larger device for a specific task, often with limited resources.

4. Circuits can reliably tell two states apart: 0 and 1. These states represent all kinds of data in binary.

I Do - Explain the model

All classroom devices connect to one central switch.

New devices can be connected easily [1], making the network straightforward to manage [1].

The switch is a single point of failure [1], so its failure can interrupt communication for connected devices [1].

Give advantage plus linked benefit and disadvantage plus linked effect.

We Do - Model answers

1. Multiple possible routes [1] mean traffic can use another path if a link fails [1]. More connections/hardware are needed [1], increasing cost or maintenance [1].

2. Each device has its own link to the central device [1], making fault isolation easier [1]. Devices can be added or managed centrally [1], simplifying expansion [1].

3. Mesh provides multiple paths between devices [1]. When a link fails [1], traffic may be routed along another available path [1], keeping communication possible [1].

Code First - Answer

```
devices = 4
links = devices * (devices - 1) DIV 2
print(links)
```

Expected output: 6

You Do - Self-mark

Q2: Several routes provide fault tolerance if a link fails [1].

Q3: The central switch connects the kiosks/computers [1], so its failure can interrupt network communication [1]. A spare or quickly replaceable switch, or a resilient design, can reduce the problem [1] by restoring service/reducing downtime [1].

Q4: Star devices connect to a central switch [1], are easier/cheaper to manage [1], but depend on the central device [1]. Mesh provides more than one path [1], so it can withstand some link failures [1], but uses extra connections increasing cost/complexity [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.